

Google Earth Engine (GEE) Backend

The `gee/` package serves as the geospatial processing backbone of the **AI-Powered Urban Heat Decision Support System**. It is responsible for acquiring satellite and environmental datasets, generating derived geospatial features, building AI-ready feature cubes, and exporting processed outputs for downstream machine learning and visualization modules.

Purpose of Each File

File	Purpose	Used By
<code>__init__.py</code>	Makes the <code>gee</code> directory a Python package.	Entire Project
<code>config.py</code>	Stores project-wide constants such as dataset IDs, CRS, spatial resolution, cloud thresholds, export settings, and default parameters.	All Processors
<code>logger.py</code>	Provides standardized logging, execution timing, success/error reporting, and debugging utilities.	All Modules
<code>data_fetch.py</code>	Handles Google Earth Engine initialization, dataset loading, collection filtering, validation, metadata retrieval, and common Earth Engine helper functions.	All Processors
<code>landsat.py</code>	Processes Landsat 8/9 imagery to generate Land Surface Temperature (LST), surface reflectance, thermal products, vegetation indices, albedo, and AI-ready feature stacks.	Data Acquisition, Feature Engineering
<code>sentinel.py</code>	Processes Sentinel-2 imagery to generate vegetation and urban indices including NDVI, NDBI, NDWI, EVI, and feature stacks.	Feature Engineering
<code>terrain.py</code>	Generates terrain-related features such as elevation, slope, aspect, hillshade, Terrain Ruggedness Index (TRI), and Topographic Position Index (TPI).	Terrain Analysis
<code>climate.py</code>	Processes ERA5-Land weather datasets to extract temperature, humidity, wind, pressure, rainfall, solar radiation, and climate-related features.	Climate Feature Engineering
<code>population.py</code>	Processes WorldPop and GHSL datasets to generate population density, exposure, and demographic features.	Human Activity Analysis
<code>urban.py</code>	Generates urban morphology features including built-up density, impervious surface, road density, building density, night-time lights, and urban feature stacks.	Urban Feature Engineering

File	Purpose	Used By
<code>pipeline.py</code>	Master orchestration engine that executes every processor, generates all derived layers, and builds the final AI-ready Feature Cube.	Entire Backend
<code>export.py</code>	Handles exporting images, feature collections, metadata, and complete Feature Cubes from Google Earth Engine to Google Drive.	Data Export
<code>utils.py</code>	Provides reusable helper utilities including geometry creation, JSON utilities, image manipulation, Earth Engine helpers, projection utilities, and common file operations.	Shared Utilities

GEE Backend Capabilities

The completed GEE package currently provides the following capabilities:

Capability	Status
Google Earth Engine Authentication	✓
Dataset Retrieval	✓
Satellite Image Processing	✓
Terrain Processing	✓
Weather Processing	✓
Population Processing	✓
Urban Feature Processing	✓
AI-Ready Feature Cube Generation	✓
Export Management	✓
Shared Utility Library	✓
End-to-End Processing Pipeline	✓

Pipeline Execution

The complete geospatial processing pipeline can be executed using:

```

from gee.pipeline import UrbanHeatPipeline

pipeline = UrbanHeatPipeline(
    region,
    start_date,
    end_date
)

feature_cube = pipeline.run()

```

The pipeline automatically:

1. Loads all required datasets.
2. Performs preprocessing.
3. Generates all geospatial features.
4. Creates the AI-ready Feature Cube.
5. Returns the complete collection of processed layers.

AI-Ready Feature Cube

The generated Feature Cube contains multiple geospatial layers that serve as inputs for downstream machine learning models.

Output Layer	Purpose
Land Surface Temperature (LST)	Primary Machine Learning Target Variable
NDVI	Vegetation Health
NDWI	Water Presence
NDBI	Built-up Area Detection
EVI	Enhanced Vegetation Health
Albedo	Surface Reflectivity
Elevation	Terrain Feature
Slope	Terrain Feature
Aspect	Terrain Orientation
Population Density	Human Exposure
Urban Density	Urban Morphology
Road Density	Infrastructure Feature

Output Layer	Purpose
Building Density	Infrastructure Feature
Impervious Surface	Urban Heat Driver
Night-time Lights	Human Activity Proxy
Weather Variables	Climate Drivers

Role of the GEE Package Within the Project

The Google Earth Engine backend forms the **foundation** of the complete Urban Heat Decision Support System.

Its responsibilities include:

- Collecting satellite imagery and environmental datasets.
- Processing multi-source geospatial data.
- Generating derived environmental and urban indicators.
- Building AI-ready raster feature stacks.
- Producing Feature Cubes for machine learning.
- Exporting processed datasets for visualization and analysis.

All subsequent modules—including Data Preprocessing, Feature Engineering, AI/ML Prediction, Heat Mapping, Explainable AI, Recommendation Engine, GIS Dashboard, and Report Generation—consume the outputs generated by this backend.

Current Backend Status

Component	Status
Configuration	✓ Complete
Logging Framework	✓ Complete
Dataset Management	✓ Complete
Landsat Processing	✓ Complete
Sentinel Processing	✓ Complete
Terrain Processing	✓ Complete
Climate Processing	✓ Complete
Population Processing	✓ Complete

Component	Status
Urban Feature Processing	✓ Complete
Processing Pipeline	✓ Complete
Export Framework	✓ Complete
Utility Library	✓ Complete

Overall Backend Completion: 100%

The `gee/` package is now a modular, extensible, and production-ready geospatial processing framework that serves as the core data acquisition and feature generation engine for the AI-Powered Urban Heat Decision Support System.